MIT Art Design and Technology University

MIT School of Computing, Pune

Cloud Storage

Class - L.Y. (SEM-I), BDCE

Unit - III

STORAGE RELIABILITY TECHNOLOGY AND STORAGE VIRTUALIZATION

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Unit III - Syllabus

MIT-ADT UNIVERSITY
PUNE, INDIA

Unit III – STORAGE RELIABILITY TECHNOLOGY AND STORAGE VIRTUALIZATION **09 hours**

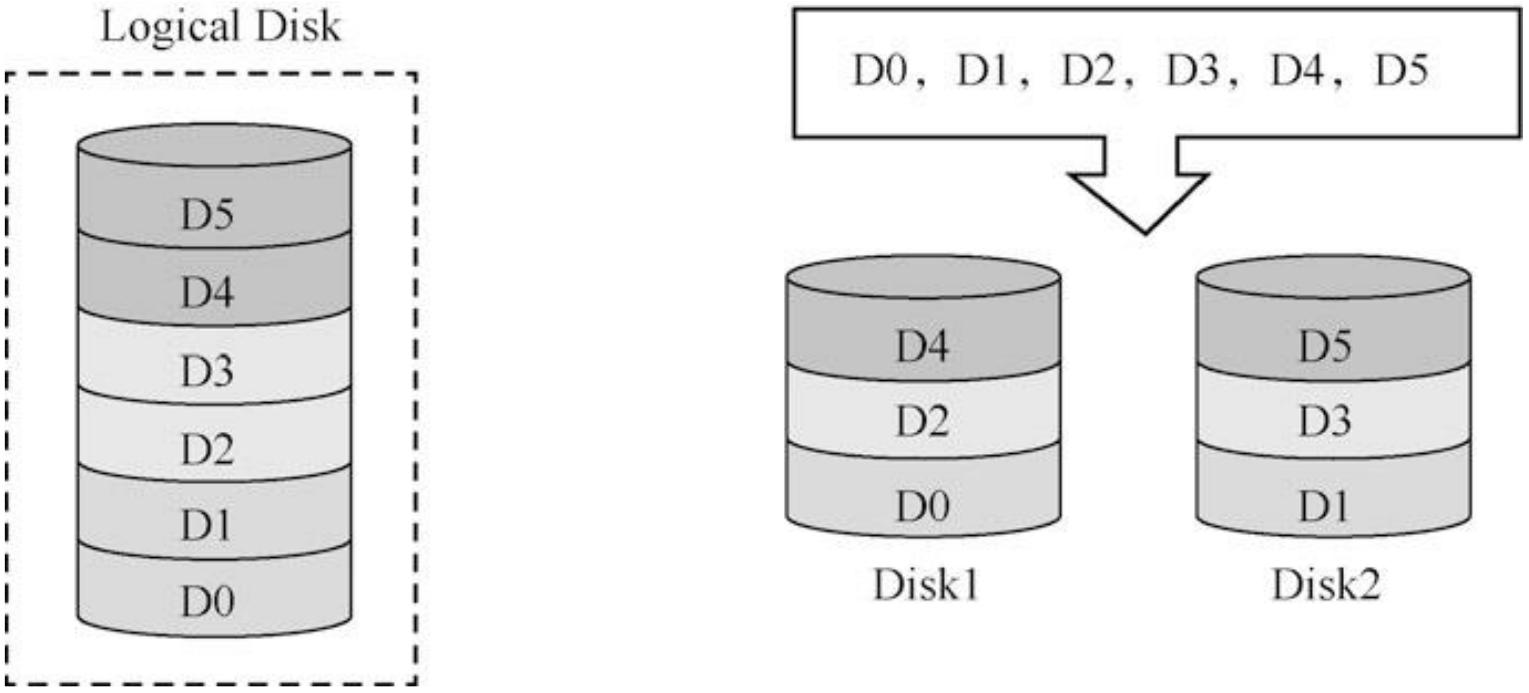
Storage Reliability Technology, Traditional RAID Technology, RAID 2.0 + Technology, Storage Virtualization, Virtualization of I/O Paths, Block-Level and File-Level Storage Virtualization, Host-Based Storage Virtualization, Storage Virtualization Based on Storage Devices, Network-Based Storage Virtualization, Storage Virtualization Products and Applications. Virtual, Physical, and Cloud Data Protection.

Storage Reliability Technology

- Traditional RAID Technology
- RAID technology has been continuously developed, now has RAID 0 to RAID 6
- total of 7 basic RAID levels. In addition, there are some basic RAID-level combinations
- such as RAID 10 (a combination of RAID 1 and RAID 0) and RAID 50 (a combination of RAID 5 and RAID 0).
- Different RAID levels represent different storage performance, data security, and storage costs.

Storage Reliability Technology

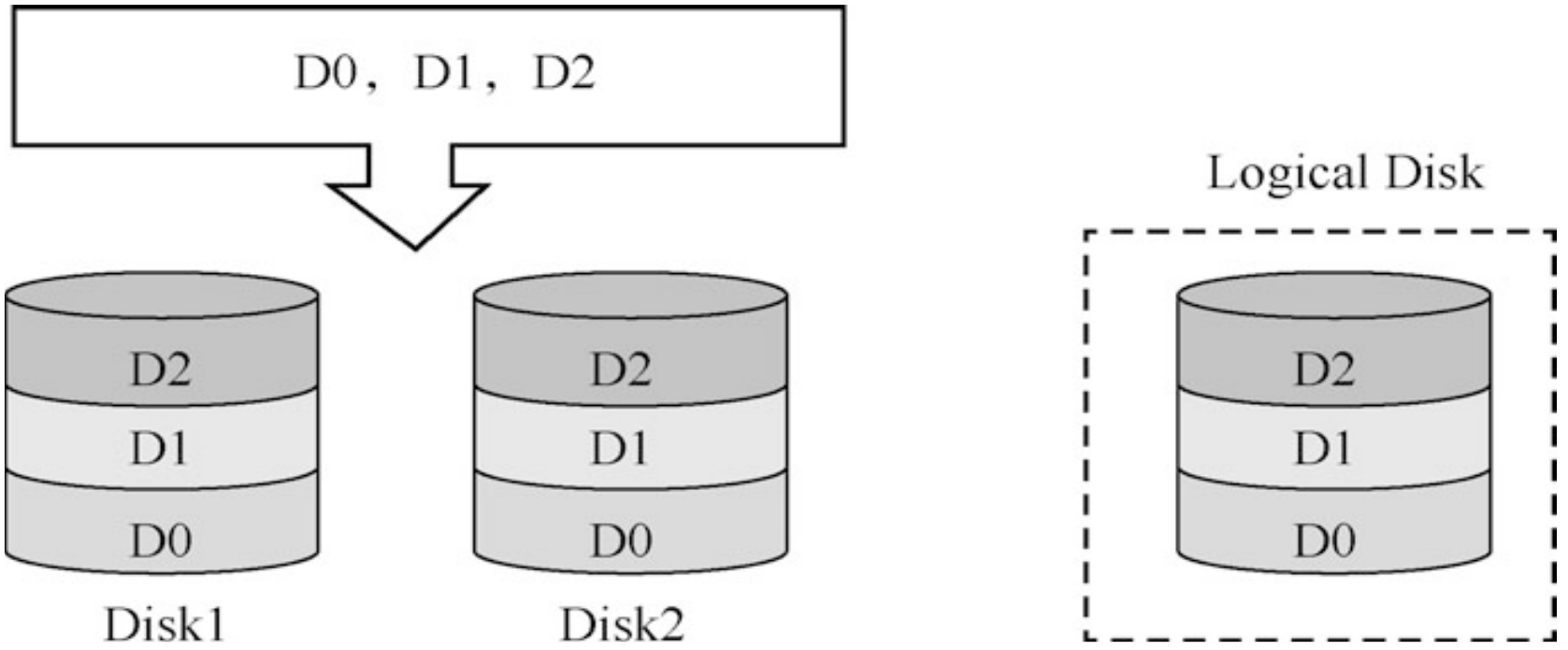
- RAID 0



- RAID 0, also known as Stripe, is based on combining multiple physical disks into a large logical disk.
- It represents the most efficient storage performance of all RAID levels, is not redundant, cannot be parallel I/O, but is the fastest.

Storage Reliability Technology

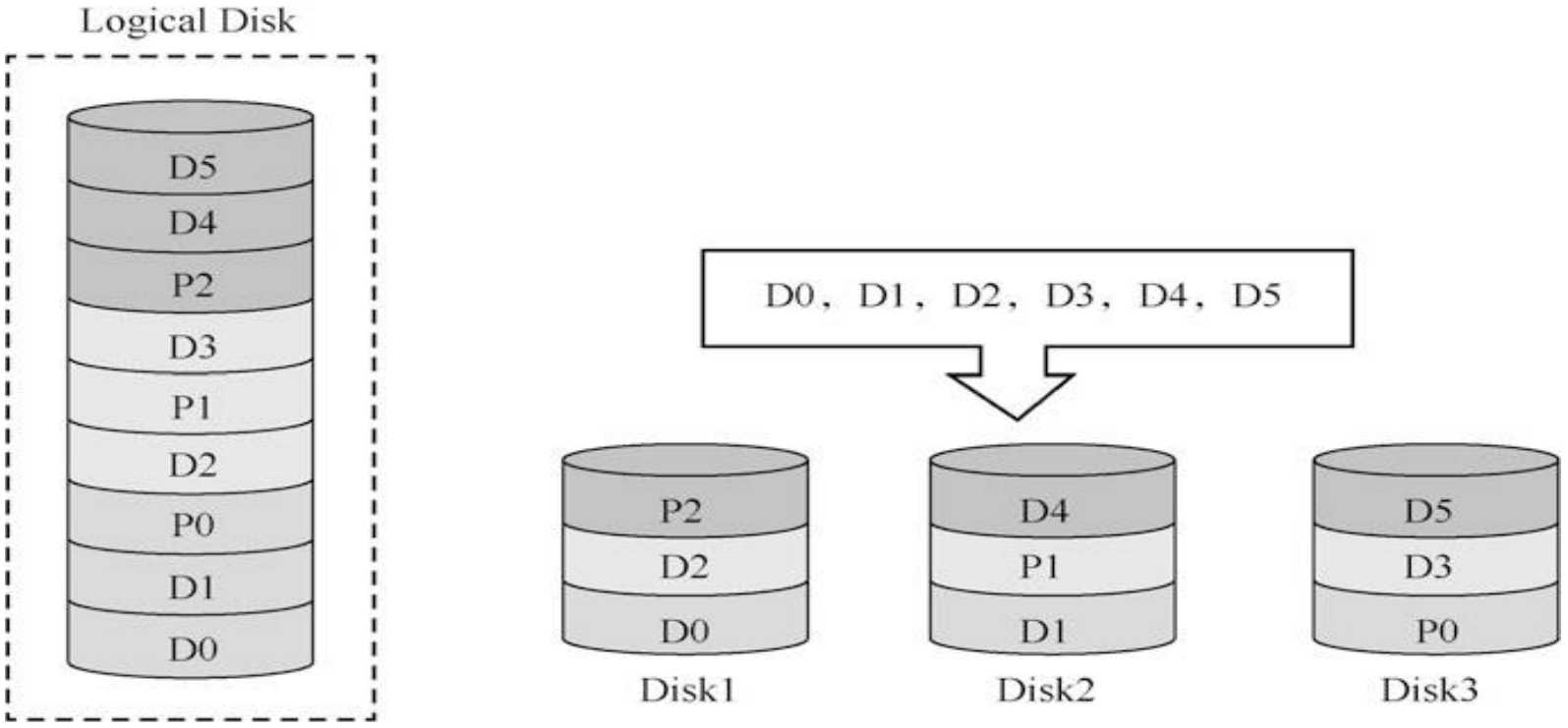
- RAID 1



- RAID 1 is also called Mirror or Mirroring. Its purpose is to maximize the availability and repairability of user data.
- The principle of RAID 1 is to automatically copy 100% of the user's data to the disk to another disk. While RAID 1 stores data on the primary disk, it also stores the same data on the mirror disk.

Storage Reliability Technology

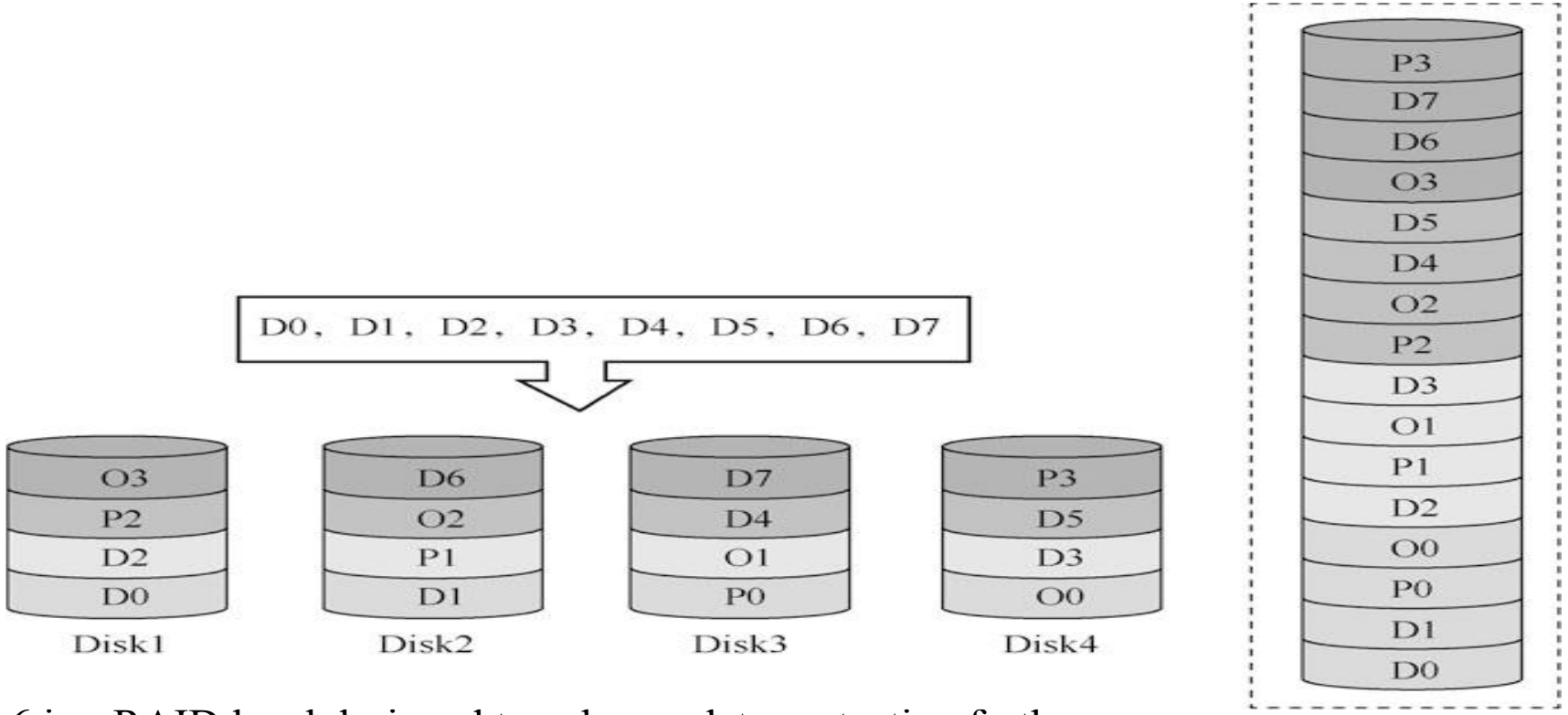
- RAID 5



- RAID 5, whose full name is “Independent Data Disk and Distributed Check Block,” is one of the most common RAID levels in advanced RAID systems.
- widely used due to its excellent performance and data redundancy balance design.
- RAID 5 uses parity for checking and error correction.

Storage Reliability Technology

- RAID 6



- RAID 6 is a RAID level designed to enhance data protection further.
- RAID 6 adds a second independent parity block compared to RAID 5. RAID 6's data security is higher than RAID 5.
- However, RAID 6 controller design is more complex, write speed is not very fast, and computational verification information and verification of data correctness takes more time.

RAID 2.0 + technology

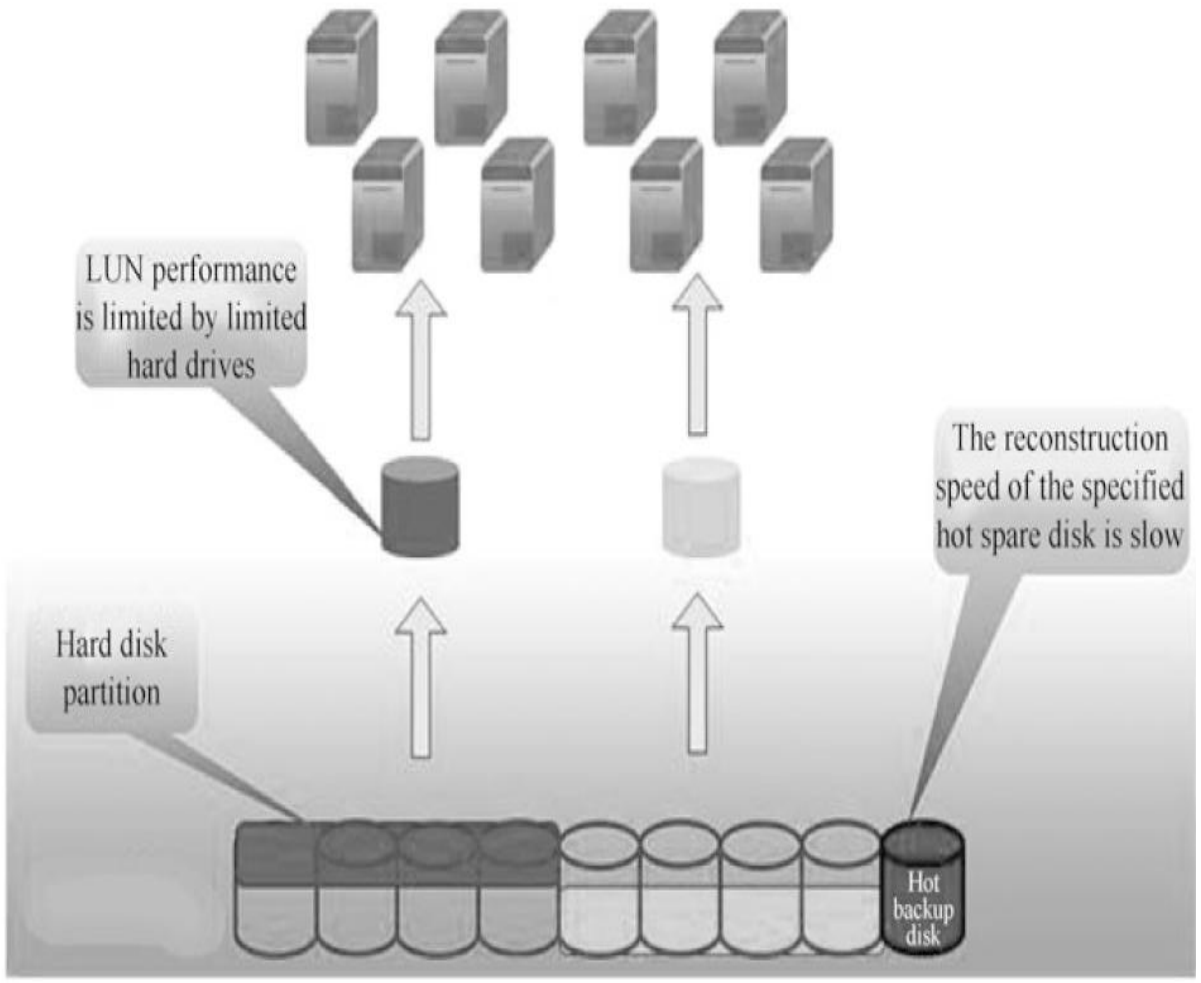


Fig. 5.23 Storage array failure recovery mechanism based on traditional RAID technology

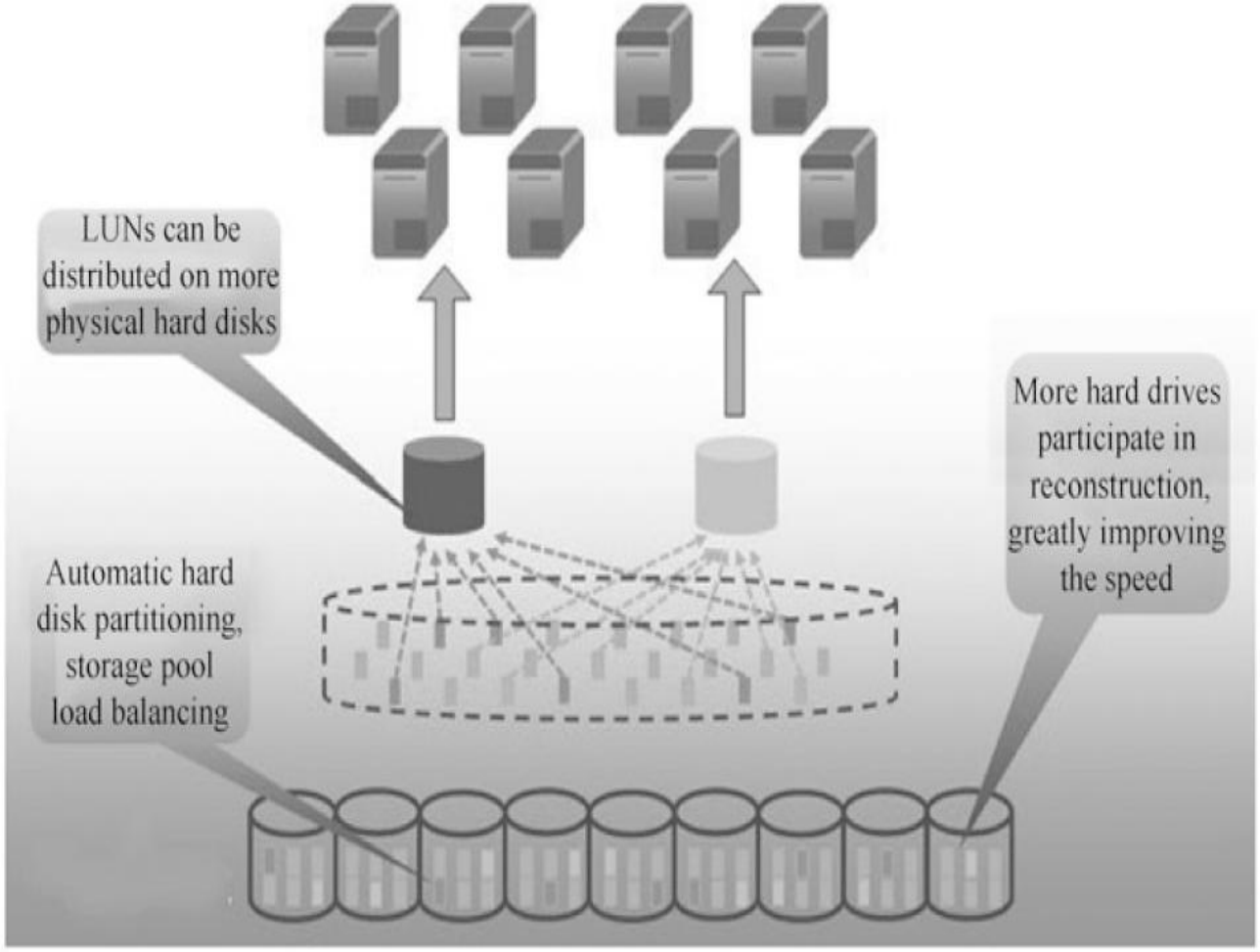


Fig. 5.24 Storage array failure recovery mechanism based on RAID 2.0 technology

RAID 2.0 + technology

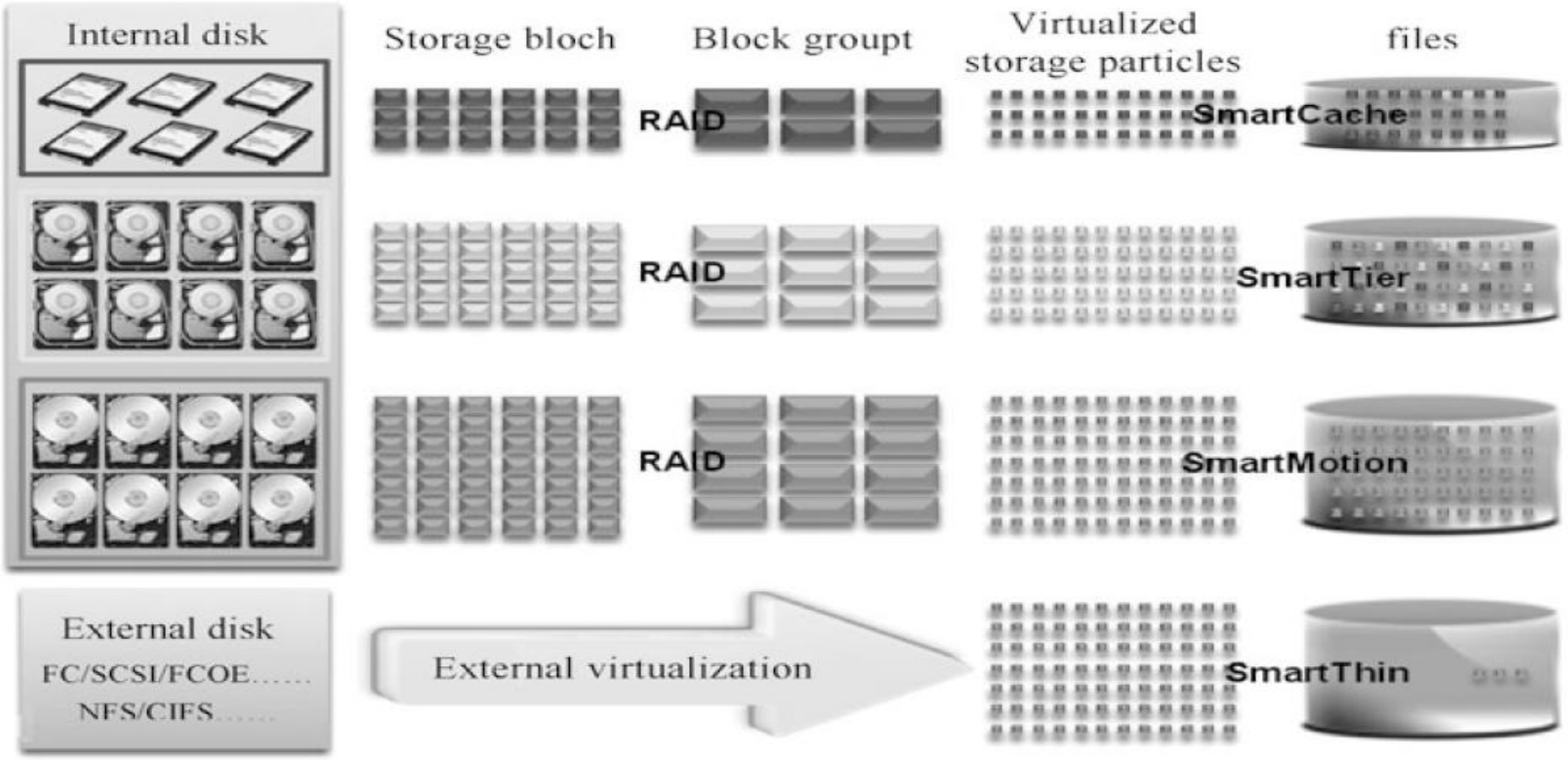


Fig. 5.25 Storage framework based on RAID 2.0+ technology

Storage Virtualization

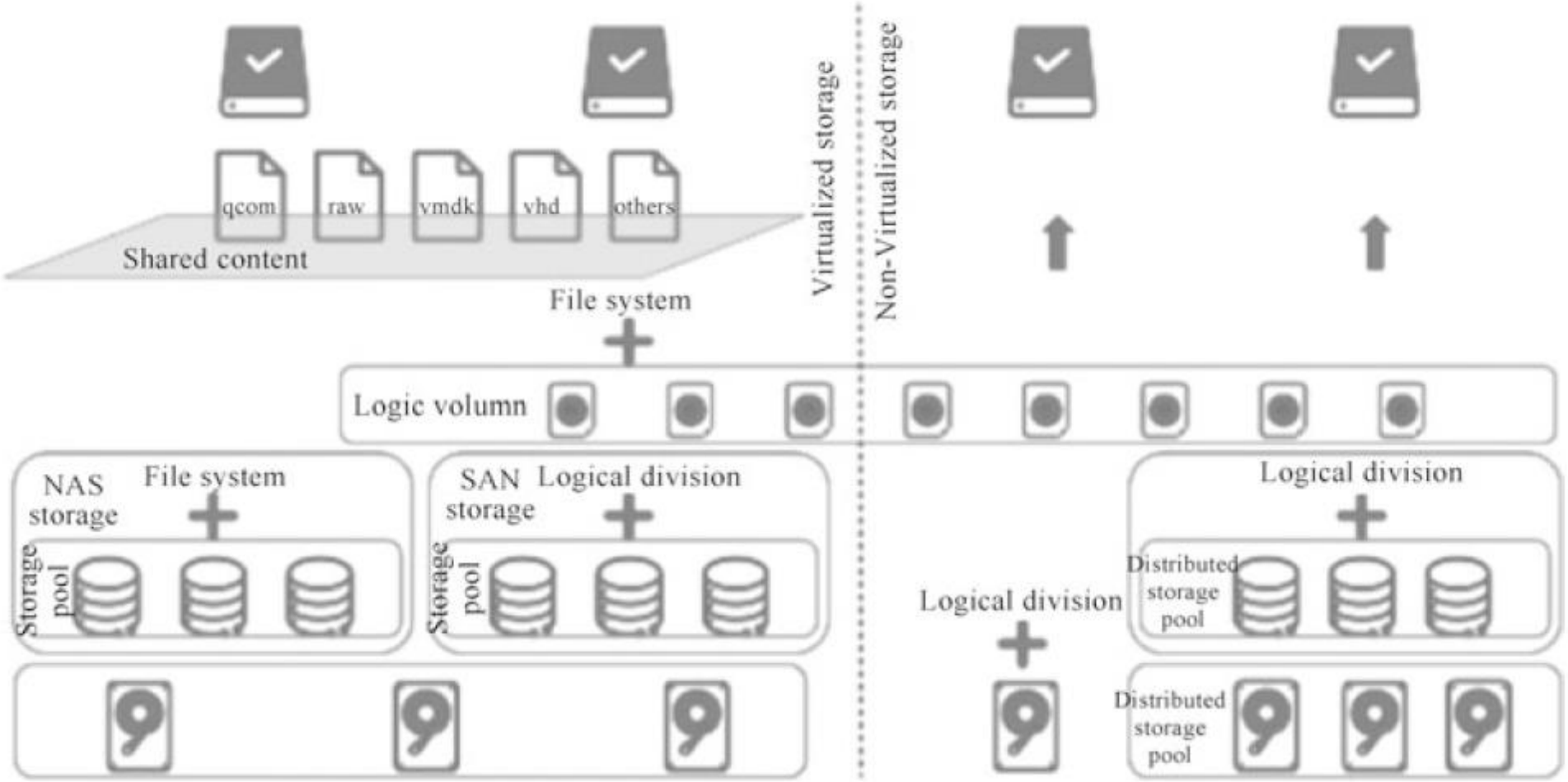


Fig. 5.29 Virtualized storage architecture

Storage Virtualization

- Storage virtualization transforms traditional computer hardware data storage into virtual storage.
- It integrates a single function within the system, enhancing the system's comprehensiveness.
- Storage virtualization technology abstracts complex and diverse real-world technology devices, allowing for reasonable control over physical devices.
- From the consumer's standpoint, storage virtualization eliminates the need to consider storage location and security, as all data is stored through virtualization.
- For enterprise managers, storage virtualization stores all data in virtualized storage pools, facilitating quicker and more efficient information management.

virtualization of i/o paths

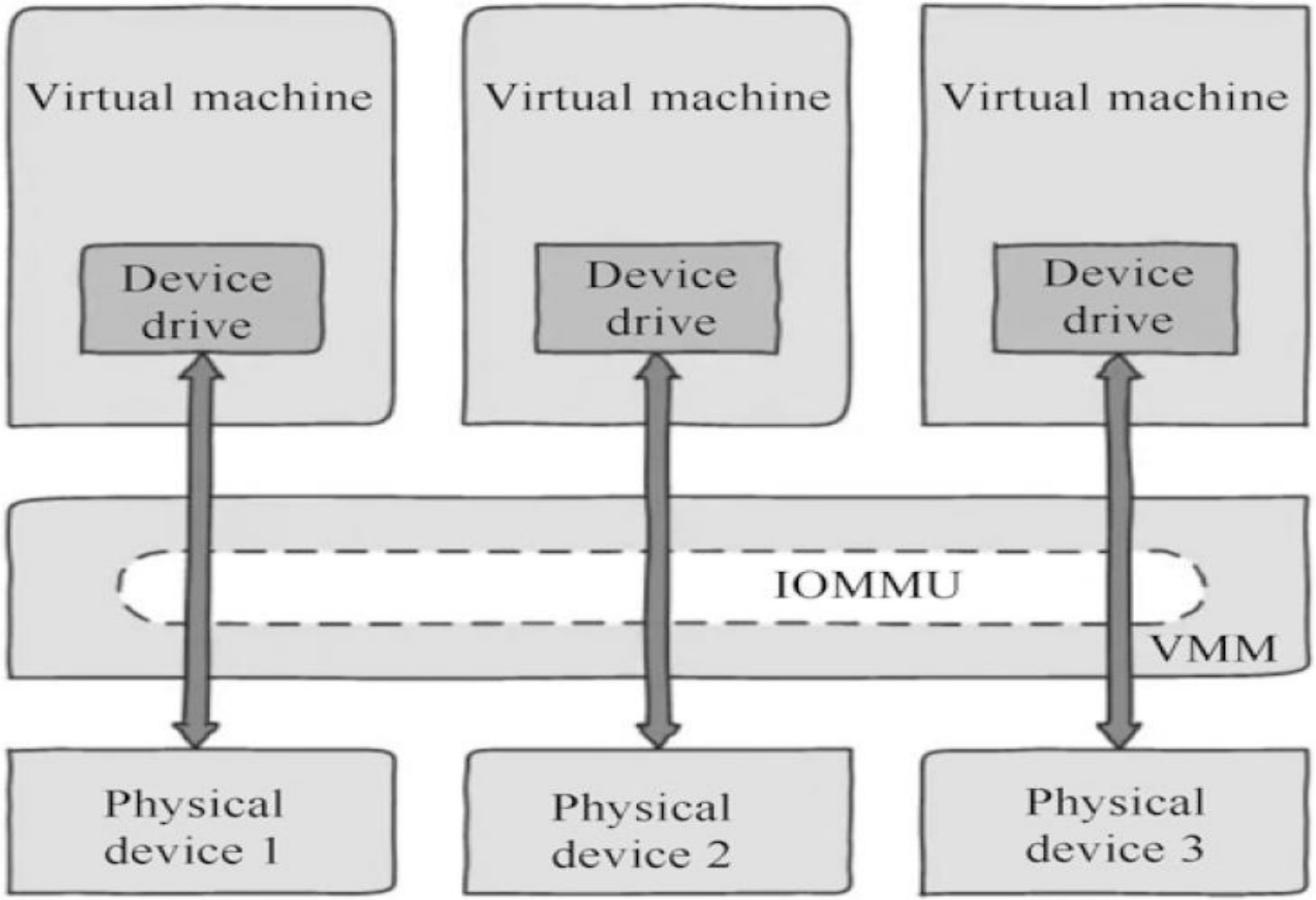


Fig. 5.30 IOMMU architecture

virtualization of i/o paths

- I/O virtualization encompasses both software-assisted and hardware-assisted virtualization methods.
- Software-assisted virtualization includes full virtualization and semi-virtualization, while hardware-assisted virtualization typically offers superior performance.
- Software-assisted virtualization can be further categorized based on device type into character device, block device, and network device I/O virtualization.
- Hardware-based virtualization, particularly I/O pass-through technology, allows virtual machines direct access to physical devices without going through a virtual machine monitor (VMM).
- I/O pass-through technology requires hardware support, such as VT-d technology, which provides significant performance enhancements without the need for operating system modifications.

Block-Level and File-Level Storage Virtualization

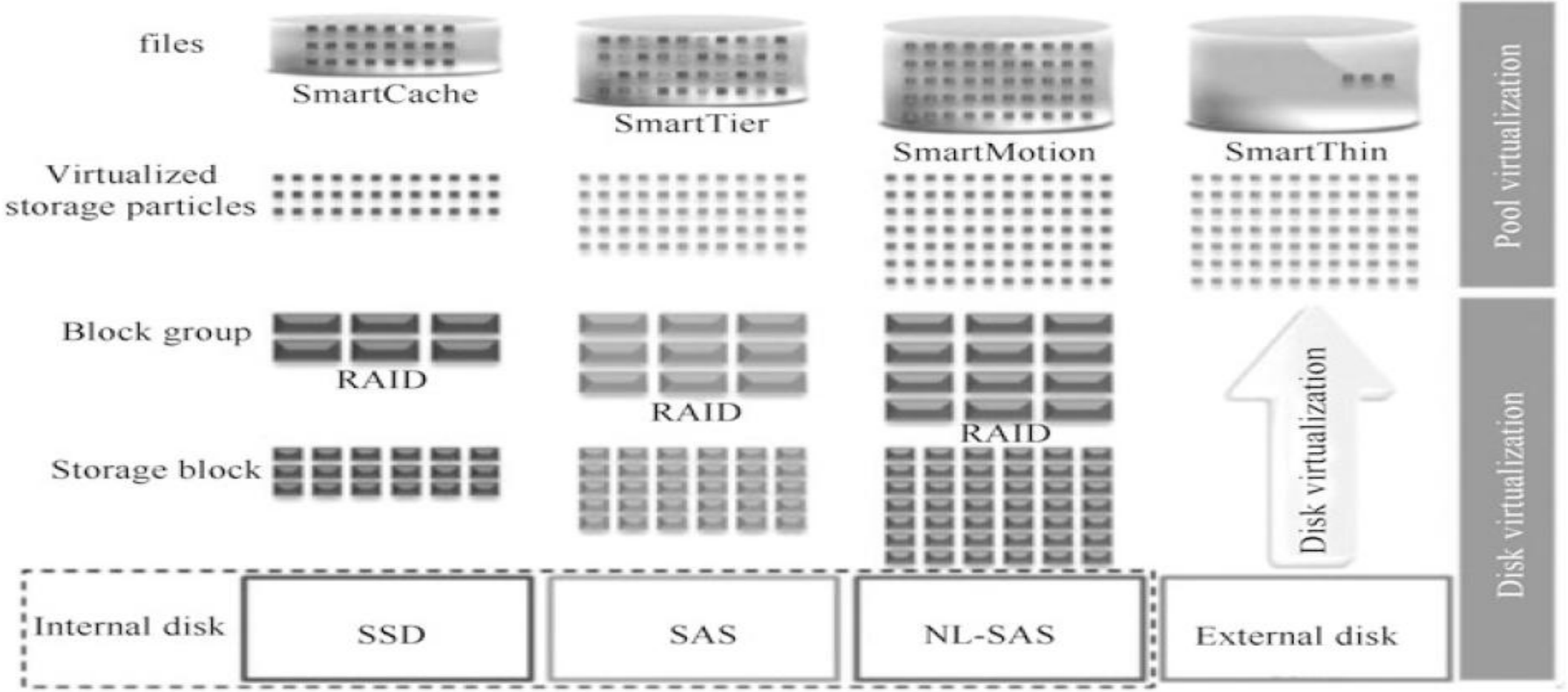


Fig. 5.32 Logical structure of block-level storage virtualization

Block-Level and File-Level Storage Virtualization

- Block-level storage virtualization abstracts physical storage addresses, providing a seamless storage service to users' programs.
- This technology enables administrators to offer scalable storage capacity without users needing to manage storage extensions or system maintenance.
- Many storage vendors, including IBM, Dell, and HP, have adopted block-level storage virtualization in their product lines through acquisitions or self-study.
- Block-level storage virtualization breaks a single disk into numerous small chunks, distributing data randomly and evenly across these chunks.
- The process involves forming storage pools, dividing hard disk space into logical blocks, grouping logical blocks into RAID groups, and creating volumes or files from fine-grained logical blocks as needed.

File-level storage virtualization

- Common file systems include NAS Storage File Systems (such as NFS and CIFS) as well as operating system file systems.
- A file system is a hierarchical organizational structure comprising many files, enabling actions like copying, pasting, deleting, and recovery.
- The organization of data into layers is determined by how the directory is structured, with the file pointer stored within the directory.
- Operating systems maintain native directories, while clusters require maintenance of NAS or shared directories created by cluster file systems.
- Examples of common operating system file formats include File Allocation Table 32-bit(FAT32) Windows, New Technology File System(NTFS) Windows, Unix File System(UFS), and Extended File System(EXT2/3/4) Linux.

File-level storage virtualization

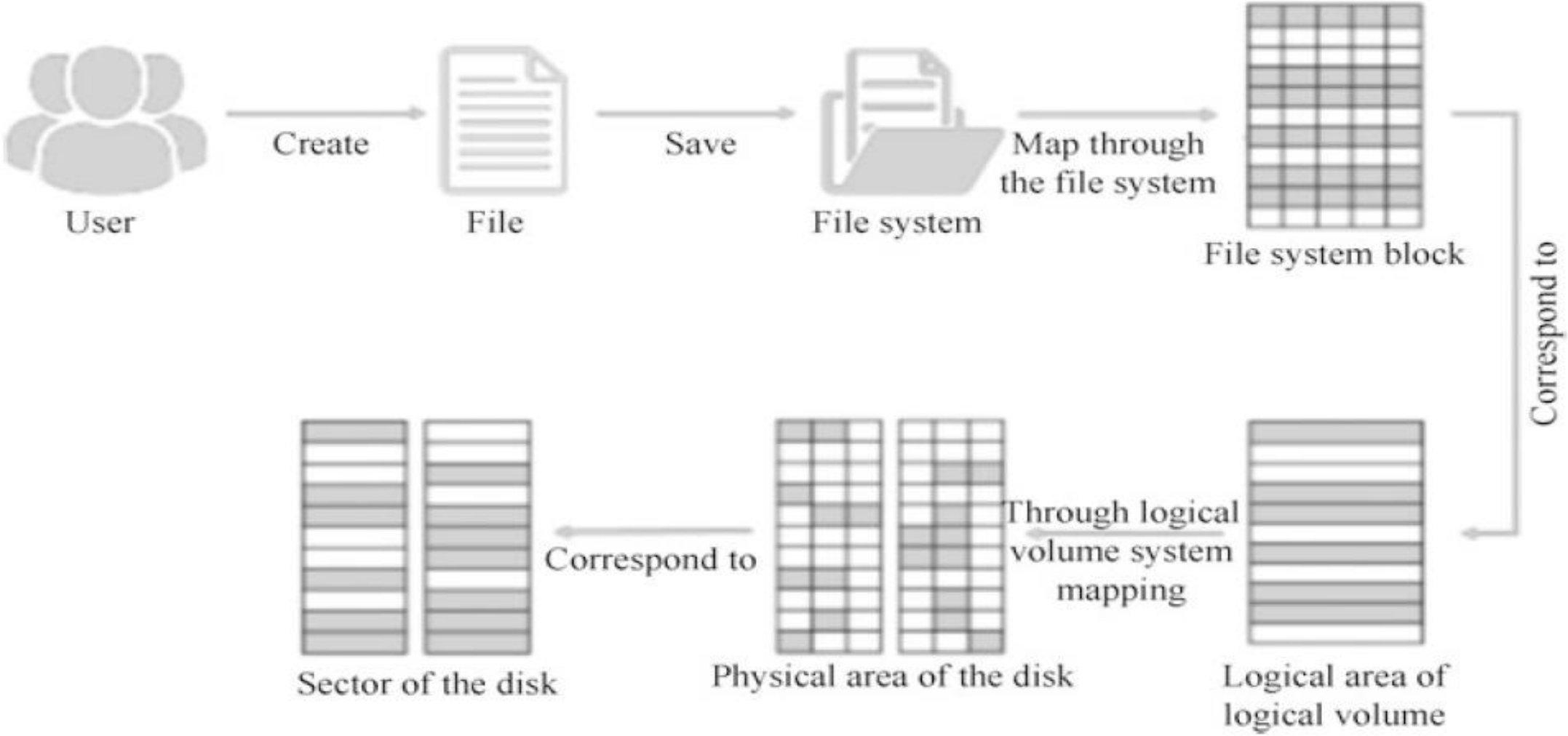


Fig. 5.35 Working process of the operating system file system

File-level storage virtualization

- Figure 5.35 shows the working process of the operating system file system. These include:
- The user or application initiates the creation of a file or folder.
- These files or folders are stored within the file system.
- The file system maps the data associated with these files to the file system block.
- The file system block corresponds to the logical region formed by the logical volume.
- The logical region is mapped to the physical area of the physical disk through the operating system or virtual machine, aligning the logical volume with the physical volume.
- The physical volume may comprise one or more physical disks within the physical region.

File-level storage virtualization

- Both centralized and distributed storage systems, utilizing RAID or replica mechanisms, result in physical volumes.
- However, for data security reasons, the entire physical volume is typically not directly mounted to upper-level applications.
- Instead, physical volumes are grouped into volume groups and further divided into logical volumes, which are then utilized by upper-layer applications.
- In cloud computing, virtualization programs format logical volumes, and various vendors employ different virtualized file systems.
- Examples include VMware's Virtual Machine File System (VMFS) and Huawei's virtual mirror management systems, both of which are high-performance cluster file systems facilitating access to a consolidated storage pool by multiple compute nodes simultaneously.

Host-Based Storage Virtualization

- Storage virtualization is a prominent direction in information storage technology, offering various implementation methods and widespread practical use.
- Virtual storage based on computer hosting utilizes volume management software embedded within common operating systems like Windows or Linux.
- Most operating systems come equipped with volume management software, eliminating the need for additional commercial software purchases for host-based storage virtualization.
- Host-based storage virtualization is cost-effective as it leverages existing software, making deployment more affordable than purchasing commercial products.
- Combining the virtual layer and file system on the same host server allows for flexible storage capacity management, dynamic adjustments without downtime, and support for heterogeneous storage systems.

Host-Based Storage Virtualization

- Drawbacks of host-based storage virtualization include reduced performance and storage expansion capabilities due to the consumption of host resources by virtual volume management software.
- Complex host upgrades, maintenance, and expansion processes can hinder business continuity, especially when dealing with multiple heterogeneous storage systems.
- Despite its drawbacks, host-based storage virtualization is relatively simple and convenient to implement, requiring only the installation of virtual volume management software on the host.
- Many virtual volume management software solutions come pre-installed with the operating system, making deployment straightforward and suitable for smaller user bases.
- Linux Logical Volume Manager (LVM) is the most commonly used product in the host-based storage virtualization market due to its widespread adoption and ease of use.

Network-Based Storage Virtualization

- Storage virtualization based on network-based storage involves adding a virtualization engine to the Storage Area Network (SAN), such as a virtual gateway, to enable virtual network storage.
- With a virtual gateway, SANs can establish storage volumes with different capacities in the virtual storage pool and manage stored data virtually.
- Network-based storage virtualization aims at consolidating heterogeneous storage systems and unifying data management.
- Advantages include independence from the host, support for heterogeneous hosts and storage devices, unified data management functions across different devices, and scalability.
- However, drawbacks include weak data management functions in some manufacturers' products, making it difficult to achieve unified data management through virtualization.

Network-Based Storage Virtualization

- **Internet device virtualization**
  - This type of virtualization relies on the symmetry of the approach to control the storage path for information and data.
  - Due to the diversity and complexity of Internet devices, achieving symmetrical storage virtualization is challenging.
  - Asymmetric storage virtualization methods are often used for Internet devices due to the separation of information control and data collection paths.
  - Asymmetric methods offer easier implementation and scalable performance compared to symmetrical methods.

- **Router device virtualization**
  - In router device virtualization, storage virtualization capabilities are integrated into router software.
  - This approach can be combined with host-based storage virtualization methods.
  - The router is placed in the host storage network channel, allowing it to acquire and process network storage commands issued by the host.
  - The router acts as the service provider for the host and manages real data virtualization storage for the computer system.
  - Router device virtualization is more independent, stable, and less disruptive compared to host-based and Internet device virtualization.
  - It may provide access to host-protected data, but only for hosts connected to the router.

Storage Virtualization Products and Applications

- **Storage Resources in FusionCompute:**
- - FusionCompute utilizes storage resources from local disks or dedicated storage devices.
- - Dedicated storage devices are connected to the host via network cable or fiber optic.
- - Storage resources are encapsulated as data stores and associated with the host.
- - These data stores can be further segmented into virtual disks for use by virtual machines.

- **Types of Encapsulated Storage Units:**
- - LUNs from SAN storage (SCSI or Fibre Channel)
- - NAS storage file systems
- - Storage pools on FusionStorage Block
- - Host's local hard drive (virtualization)

- **Management of Storage Resources:**
- - Storage units are collectively referred to as "storage devices" in Huawei FusionCompute.
- - Physical storage media providing storage space is termed "storage resources."
- - Storage resources must be manually added before utilization.
- - For IP-SAN, FusionStorage, or NAS storage, a storage interface needs to be added to hosts for communication with centralized storage controllers.

Virtual, Physical, and Cloud Data Protection.

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- **Virtual Data Protection:**
  - Involves safeguarding data stored within virtual environments, such as virtual machines (VMs) or virtual disks.
  - Utilizes backup and recovery solutions specifically designed for virtualized infrastructures.
  - Offers features like snapshot-based backups, agentless backup solutions, and VM-level backup and recovery.
  - Ensures data integrity and availability within virtualized environments, minimizing downtime and data loss.

- **Physical Data Protection:**
  - Focuses on securing data stored on physical servers, storage devices, and endpoints.
  - Involves traditional backup and recovery methods like tape backups, disk imaging, and file-level backups.
  - Requires dedicated backup agents installed on each physical device to facilitate data backup and restore operations.
  - Often includes disaster recovery planning to mitigate risks associated with hardware failures, natural disasters, or other physical threats.

- **Cloud Data Protection:**
  - Addresses the security and resilience of data stored in cloud environments, including public, private, and hybrid clouds.
  - Incorporates cloud-native backup and recovery services provided by cloud service providers (CSPs).
  - Emphasizes encryption, access controls, and compliance measures to protect data in transit and at rest within cloud infrastructures.
  - Integrates with cloud-native tools and APIs to automate data backup, replication, and recovery processes.

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Thank You!!!